6a	Each coil acts as a current-carrying conductor in the magnetic field of the neighbouring turn Magnetic field (due to current) in one loop cuts / is normal to the current in second loop. By Fleming's left-hand rule, there will be a (magnetic) force on the second loop towards the first loop. (By Newton's third law, this gives rise to mutual attraction between the loops.) Hence, separation decreases.	**A**	 M1 M1 A0
6bi	Mass exert a downward force on the frame. / mass exert a (clockwise) moment on the frame For frame to have no net moment about pivot (or for the frame to be in rotational equilibrium**)**, magnetic force must act downwards on near side of frame.	**A**	B1 A1
6bii	$F = BIL$ $24 \times 10^{-3} = B \times 4.5 \times 4.6 \times 10^{-2}$ $B = 0.116\text{ T} = 0.12\text{ T}$	**A**	 C1 A1
7a	When the rate of decay of Xe-143 is faster than that of Cs-143, the number of Cs-143	**A**	B1